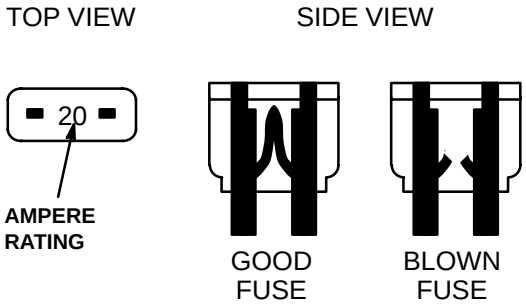
11-1 FUSE PANEL/CIRCUIT PROTECTION

CIRCUIT PROTECTION DEVICES

Electrical circuits on this vehicle may be protected by fuses, fusible links, maxi-fuses, circuit breakers, or a combination of these devices.

BLADE TYPE FUSE

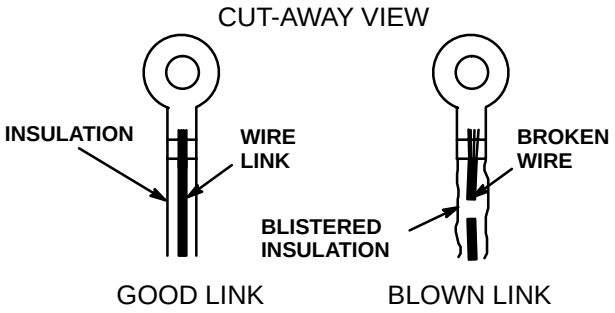


Blade type fuses have a transparent plastic housing. To check a fuse, pull it from the fuse panel and look at the fuse element through the housing. Always replace a blown fuse with a new fuse that has the same ampere rating.

The ampere rating of a blade type fuse can also be determined by following the color code shown here:

BLADE FUSE COLOR CODING	
AMPERE RATING	**HOUSING COLOR**
4	Pink
5	Tan
10	Red
15	Light Blue
20	Yellow
25	Natural
30	Light Green

FUSIBLE LINK



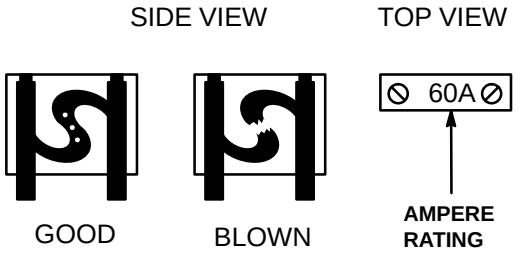
Fusible links are short lengths of wire that are smaller in diameter than the wires they are protecting. Fusible link wire is covered with a special thick, non-flammable insulation. An overload condition causes the insulation to blister. If the overload condition continues, the wire link will melt. To check a fusible link, look for blistered insulation. If the insulation is okay, pull lightly on the wire; if the fusible link stretches, the wire has melted.

When replacing fusible links, first cut the protected wire where it is connected to the fusible link. Then, tightly crimp or solder the new link to the protected wire.

Fusible links are often identified by color coding of the insulation, as shown here:

FUSIBLE LINK COLOR CODING	
WIRE LINK SIZE	**INSULATION COLOR**
20 GA	Blue
18 GA	Brown or Red
16 GA	Black or Orange
14 GA	Green
12 GA	Gray

MAXI-FUSE



Maxi-fuses have a transparent colored plastic housing. To check a maxi-fuse, look at the fuse element through the side of the housing.

To replace a maxi-fuse, pull it from the fuse box or panel. Always replace a blown maxi-fuse with a new one having the same ampere rating.

The ampere rating of a maxi-fuse can also be determined by following the color code shown here:

MAXI-FUSE COLOR CODING	
AMPERE RATING	**HOUSING COLOR**
30	Light Green
40	Amber
50	Red
60	Blue

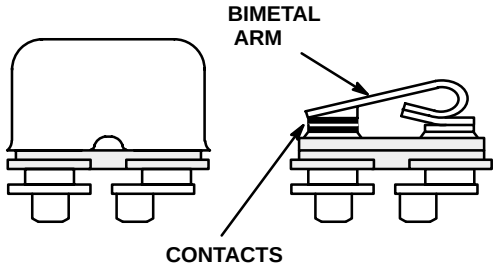
CIRCUIT BREAKER

Some circuits are protected by circuit breakers (abbreviated “c. b.” in fuse chart).
They can be Fuse Panel mounted or in-line. Like fuses, they are rated in amperes.

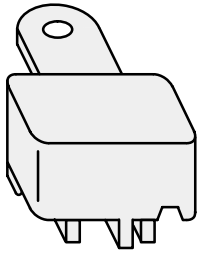
Each circuit breaker conducts current through an arm made of two types of metal bonded together (bimetal arm). If the arm starts to carry too much current, it heats up. As one metal expands faster than the other, the arm bends, the contacts open, and current flow is broken. A circuit breaker can be the cycling or non-cycling type.

FUSE PANEL MOUNTED CYCLING TYPE

SIDE VIEW (EXTERNAL) SIDE VIEW (INTERNAL)



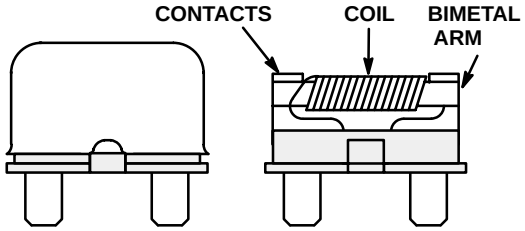
IN-LINE MOUNTED CYCLING TYPE



In the cycling type, the bimetal arm cools and straightens out. This cycle repeats as long as the overcurrent exists and power is applied.

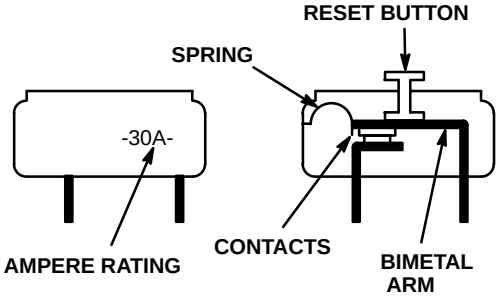
FUSE PANEL MOUNTED NON-CYCLING TYPE

SIDE VIEW (EXTERNAL) SIDE VIEW (INTERNAL)



FUSE PANEL MOUNTED MANUAL RESET TYPE

SIDE VIEW (EXTERNAL) SIDE VIEW (INTERNAL)



Two types of non-cycling circuit breakers are used; one is reset by removing power from the circuit, and the other is reset by depressing a reset button.

In the first type, there is a coil wrapped around the bimetal arm. When an overcurrent exists and the contacts open, a small current passes through the coil. This current through the coil is not enough to operate a load, but it does heat up both the coil and the bimetal arm. This keeps the arm in the open position until power is removed.

In the second type, a spring pushes the bimetal arm down and holds the contacts together. When an overcurrent condition exists and the bimetal arm heats up, the bimetal arm bends enough to overcome the spring and the contacts snap open. The contacts stay open until the reset button is pushed and the contacts snap together again.

DIODE

Diodes are electrical devices that permit current to flow in one direction only. The current flows in the direction indicated by the arrow.

MEGA FUSE

TOP VIEW

Mega fuse protects the charging circuit. In the event that the mega fuse is blown due to the charging circuit failure, the generator field circuit is disabled.

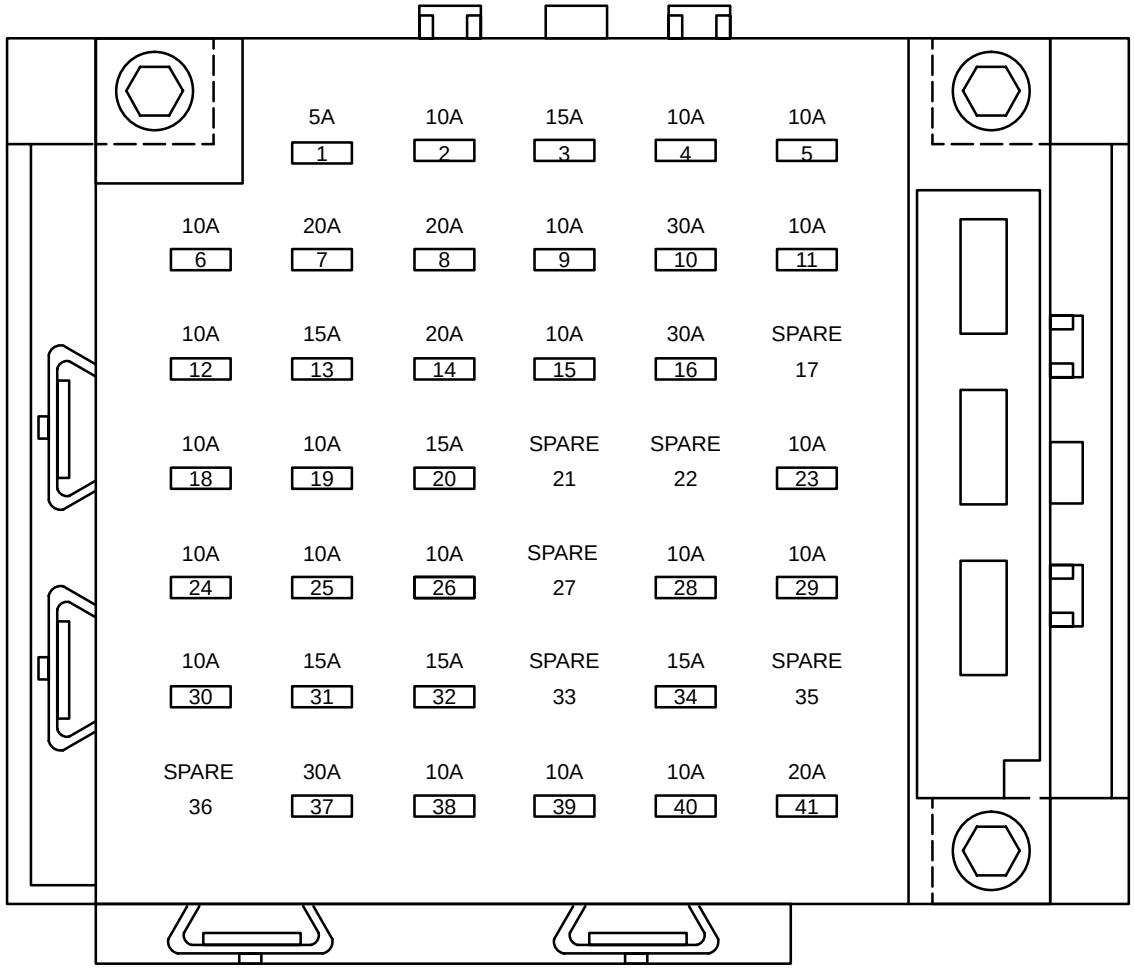
11-3 FUSE PANEL/CIRCUIT PROTECTION

Power Distribution

The Generator and Battery are connected together at the Battery Junction Box. Other circuits originate at the Battery Junction Box and are protected by fuses.

All Mega/Maxi-Fuses and Mini-Fuses in the Battery Junction Box, the Ignition Switch and the Lighting Control Module (Integrated Main Light Switch) are powered at all times. Fuses 1, 2, 7, 8, 13, 14, 19, 20, 25, 26, 31, 32, 37, 38, 39 and 41, are also powered at all times. All other fuses are powered through the Ignition Switch, and Rear Window Defrost Control.

CENTRAL JUNCTION BOX

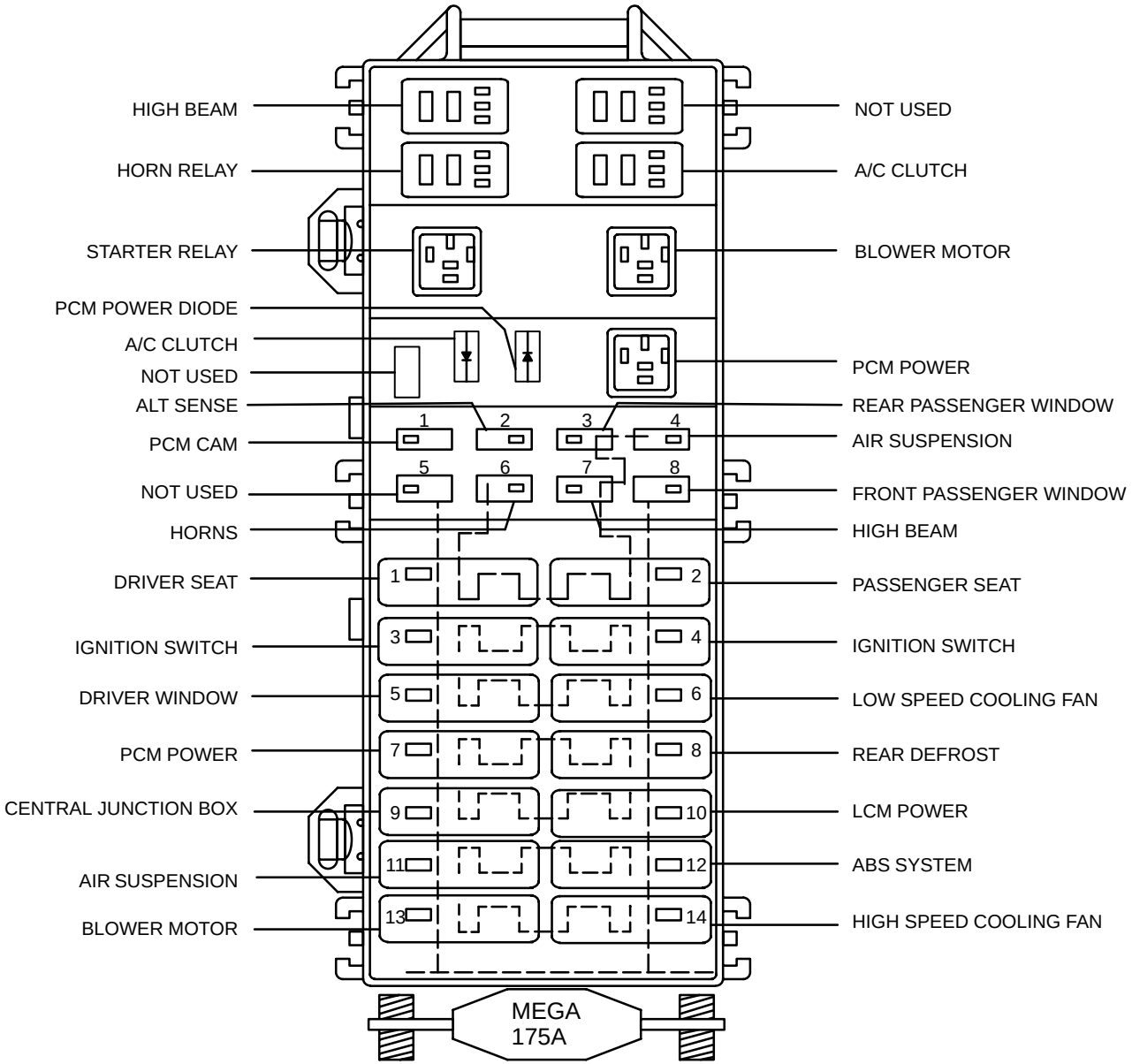


CENTRAL JUNCTION BOX

FUSE	AMPS	CIRCUITS PROTECTED
1	5	Lighting Control Module: Anti-Theft Indicator Lamp, PWM Dimming Output Illumination Lamps for Microphone, Heated Seat Switches, Rear Defrost Control Switch, EATC Control Panel, Message Center Switches, Speed Control Switches, Cigar Lighter, Console and Ashtray
2	10	Data Link Connector (DLC), Powertrain Control Module (PCM)
3	15	Multifunction Switch: Cornering Lamps, High Beam and Turn Signal Input to LCM
4	10	Power Door Locks and Power Windows Switch Backlites, Front Radio Control Unit, Cellular Telephone Transceiver, Lighting Control Module, (RUN/ACC Sense), Electronic Day/Night Mirror
5	10	Virtual Image Instrument Cluster, Lighting Control Module (LCM RUN/START Sense), Light Sensor/Amplifier
6	10	Virtual Image Instrument Cluster, RF Park/Turn Lamp
7	20	Power Point
8	20	Fuel Filler Door Release Switch, Luggage Compartment Lid Relay
9	10	Restraints Control Module, EATC Module, Blower Motor Relay
10	30	Windshield Wiper Motor, Windshield Wiper Module
11	10	Ignition Coils, Radio Interference Capacitor, PCM Power Relay, Passive Anti-Theft System (PATS) Module
12	10	Lighting Control Module (LCM)
13	15	Lighting Control Module (LCM): RF Turn Lamp, Right Turn Indicator (VIC), RR Side Lamps, Tail Lamps, License Lamps, LR Stop/Turn Lamps
14	20	Cigar Lighter
15	10	ABS EVAC and Fill Connector, RESCU Module
16	30	Roof Opening Panel Switch
17	—	NOT USED
18	10	Lighting Control Module (LCM)
19	10	Lighting Control Module (LCM): Left Headlamp, DRL
20	15	Multifunction Switch: Flash to Pass, and Hazard Warning Input to LCM
21	—	NOT USED
22	—	NOT USED
23	10	Digital Transmission Range (DTR) Sensor
24	10	Virtual Image Instrument Cluster, LF Park/Turn Lamp
25	10	Lighting Control Module (LCM): Right Headlamp
26	10	Virtual Image Instrument Cluster, EATC Module
27	—	NOT USED
28	10	Shift Lock Actuator, Vehicle Dynamic Module, Virtual Image Instrument Cluster, Rear Window Defrost, Heated Seat Switch Assembly, Low Tire Pressure Module, Speed Control Servo
29	10	Front Control Unit (Radio)
30	10	Heated Mirrors
31	15	Lighting Control Module (LCM): FCU, Electronic Day/Night Mirror, RH and LH I/P Courtesy Lamp, Door Courtesy Lamps, RH and LH Map Lamps, RR and LR Reading Lamps, RH and LH Visor Lamps, Storage Bin Lamp, Luggage Compartment Lid Lamp, Glove Box Lamp, Light Sensor Amplifier
32	15	Speed Control DEAC Switch, Brake Pedal Position (BPP) Switch
33	—	NOT USED
34	15	Console Shift Illumination, A/C Clutch Cycling Pressure Switch, A/C Clutch Relay, DTR Sensor
35	—	NOT USED
36	—	NOT USED
37	30	Subwoofer Amplifier, Front Control Unit (Radio)
38	10	Clock, CD Player, Cellular Telephone Transceiver, RESCU
39	10	Power Door Locks, Power Seats, Exterior Rear View Mirrors, Keyless Entry, LF Seat Module, LF Door Module
40	10	Multifunction Switch: Cornering Lamps
41	20	LF Door Module

11-5 FUSE PANEL/CIRCUIT PROTECTION

BATTERY JUNCTION BOX



BATTERY JUNCTION BOX

MEGA-FUSE	AMPS	CIRCUITS PROTECTED
–	175	Generator
MAXI-FUSE	**AMPS**	**CIRCUITS PROTECTED**
1	30	Driver Seat Module, LF Heated Seat Module, LF Power Lumbar
2	30	Passenger Seat Module, RF Heated Seat Module, RF Power Lumbar
3	40	Ignition Switch
4	40	Ignition Switch, Starter Relay
5	40	Rear Driver Window, LF Door Module
6	30	Low Speed Cooling Fan Relay
7	30	PCM Power
8	40	Rear Window Defrost Control
9	60	Central Junction Box
10	60	LCM Power, Power Point
11	60	Compressor Relay
12	60	Anti-Lock Brake Control Module, ABS EVAC and Fill Connector
13	40	Blower Motor Relay
14	60	High Speed Cooling Fan Relay

MINI-FUSES	AMPS	CIRCUITS PROTECTED
1	30	PCM Components, AX4N Transaxle
2	20	Generator
3	30	Rear Passenger Window
4	30	Air Suspension
5	–	NOT USED
6	20	Horn Relay
7	15	High Beam Relay
8	30	Front Passenger Window